

DEVALIVE

Product Vision Document (PVD)

Product Requirements Document (PRD)

Version: 1.0

Prepared By: Tanmay Bonde

Project Type: SaaS Application

Status: Planning Phase

1. Executive Summary

DevAlive is a SaaS-based developer uptime monitoring and project activity management platform designed to help developers keep their deployed projects active, accessible, and recruiter-ready.

Modern developers frequently host applications on free-tier cloud services such as Render, Railway, Vercel, Netlify, Supabase, Firebase, and MongoDB Atlas. Many of these platforms implement inactivity policies that suspend applications, databases, or backend services after prolonged periods of inactivity.

As a result, portfolio projects often become unavailable when recruiters, clients, interviewers, or end-users attempt to access them.

DevAlive solves this problem by continuously monitoring registered projects, performing automated health checks, tracking uptime metrics, generating monitoring analytics, and sending failure alerts whenever a project becomes unavailable.

The platform provides developers with confidence that their deployed projects remain operational and presentable at all times.

2. Product Vision

Vision Statement

To become the most developer-friendly uptime monitoring and project reliability platform that ensures every portfolio project remains active, accessible, and recruiter-ready.

3. Mission Statement

DevAlive exists to eliminate the frustration and embarrassment developers experience when their deployed projects become inactive or unavailable due to free-tier hosting limitations.

The platform empowers developers by providing:

- Automated project monitoring
- Continuous health checking
- Uptime analytics
- Downtime alerts
- Reliability insights

without requiring complex infrastructure monitoring tools.

4. Problem Statement

Developers frequently deploy projects on:

- Render
- Railway
- Vercel
- Netlify
- Firebase
- Supabase
- MongoDB Atlas

Many free-tier services automatically:

- Pause applications
- Suspend backend servers
- Disable databases
- Introduce cold starts
- Limit background execution

when projects remain inactive for extended periods.

This creates multiple issues:

For Developers

- Portfolio projects appear broken.
- Recruiters lose confidence.
- Demo applications become unavailable.
- Client prototypes stop functioning.
- Developers remain unaware of failures.

For Recruiters

- Unable to evaluate projects.
- Receive server errors.
- Experience failed application launches.
- View incomplete portfolios.

For Freelancers

- Demo links become inaccessible.
- Potential clients see unreliable products.

Existing enterprise monitoring solutions are often:

- Expensive
- Over-engineered
- Infrastructure-focused
- Difficult for beginners

Developers need a simple solution focused specifically on personal projects and portfolio applications.

5. Product Purpose

The purpose of DevAlive is to provide a centralized platform where developers can:

- Register projects
- Monitor availability
- Maintain endpoint activity
- Track uptime
- Detect downtime
- Receive alerts
- Analyze project health

through an intuitive SaaS dashboard.

6. Target Audience

Primary Audience

Student Developers

Characteristics:

- Learning deployment
- Building portfolios
- Using free-tier services
- Preparing for placements

Needs:

- Project reliability
- Recruiter-ready portfolios
- Simple setup

Junior Software Engineers

Characteristics:

- Building side projects
- Applying for jobs
- Maintaining personal portfolios

Needs:

- Monitoring
- Alerts
- Reliability tracking

Freelancers

Characteristics:

- Hosting client prototypes
- Maintaining demo applications

Needs:

- Downtime visibility

- Alert notifications
 - Project health tracking
-

Indie Hackers

Characteristics:

- Running multiple projects
- Launching MVP products

Needs:

- Availability monitoring
 - Analytics
 - Project management
-

7. Product Goals

Business Goals

- Create a developer-focused SaaS product.
 - Solve a real-world developer problem.
 - Build a scalable monitoring platform.
 - Establish foundation for future premium features.
-

User Goals

Users should be able to:

- Know whether projects are online.
 - Receive downtime alerts.
 - Track uptime.
 - Maintain portfolio reliability.
 - Access monitoring history.
-

8. Success Metrics

The product will be considered successful when:

User Metrics

- User registrations increase consistently.
- Users actively add projects.
- Users continue monitoring projects.

Platform Metrics

- Monitoring accuracy above 99%.
- Reliable alert delivery.
- Low false-positive downtime reports.

Business Metrics

- Monthly active users growth.
- Project retention rate.
- Subscription conversion rate (future).

9. Competitive Advantage

Unlike enterprise monitoring tools:

- Simpler setup
- Portfolio-focused
- Developer-centric
- Beginner-friendly
- Affordable
- SaaS dashboard experience

DevAlive focuses specifically on developers using free-tier infrastructure.

10. Product Scope

Included in MVP

- User Authentication
- Project Registration
- Endpoint Monitoring
- Monitoring Scheduler
- Monitoring Logs
- Uptime Tracking
- Analytics Dashboard
- Email Alerts
- User Dashboard

- Project Management
-

Excluded from MVP

- Team Collaboration
 - Multi-user Organizations
 - SMS Alerts
 - Push Notifications
 - Slack Integration
 - Discord Integration
 - API Monitoring Chains
 - Infrastructure Monitoring
 - AI Analytics
-

11. Product Requirements

Functional Requirements

Module 1 — Authentication

Features

- User Registration
- User Login
- User Logout
- JWT Authentication
- Password Encryption
- Session Management

User Stories

As a user:

- I want to create an account.
 - I want to log in securely.
 - I want my account protected.
 - I want access to my projects only.
-

Module 2 — Project Management

Features

- Add Project
- Edit Project
- Delete Project
- Enable Monitoring
- Disable Monitoring

User Stories

As a user:

- I want to register projects.
 - I want to update project information.
 - I want to remove projects.
 - I want monitoring control.
-

Module 3 — Monitoring Engine

Features

- Scheduled Health Checks
- HTTP Monitoring
- HTTPS Monitoring
- Failure Detection
- Retry Mechanism

User Stories

As a user:

- I want DevAlive to monitor my project automatically.
 - I want failures detected immediately.
 - I want uptime records maintained.
-

Module 4 — Monitoring Logs

Features

- Monitoring History
- Success Logs
- Failure Logs

- Timestamp Tracking

User Stories

As a user:

- I want to see previous monitoring results.
 - I want detailed monitoring records.
-

Module 5 — Analytics Dashboard

Features

- Uptime Percentage
- Downtime Statistics
- Monitoring Frequency
- Project Status Visualization

User Stories

As a user:

- I want visual analytics.
 - I want uptime insights.
 - I want reliability statistics.
-

Module 6 — Alert System

Features

- Email Notifications
- Downtime Alerts
- Monitoring Failure Alerts

User Stories

As a user:

- I want immediate downtime notifications.
 - I want to know when projects fail.
-

12. Non-Functional Requirements

Performance

- Dashboard loads within 3 seconds.
 - APIs respond within 2 seconds.
 - Monitoring executes asynchronously.
-

Reliability

- Monitoring scheduler must operate continuously.
 - Failures must be logged.
 - Recovery mechanisms should exist.
-

Security

- JWT Authentication
 - Password Hashing
 - Input Validation
 - HTTPS Communication
 - Protected Routes
 - Environment Variable Security
-

Scalability

Architecture must support future:

- Team Management
 - Enterprise Accounts
 - Multi-Region Monitoring
 - Premium Plans
-

Availability

Monitoring services should remain operational whenever backend services are active.

13. Future Roadmap

Phase 2

- Slack Alerts
 - Discord Alerts
 - Telegram Alerts
-

Phase 3

- Team Workspaces
 - Organization Accounts
 - Shared Dashboards
-

Phase 4

- AI Monitoring Insights
 - Failure Prediction
 - Automated Recommendations
-

Phase 5

- Multi-Region Monitoring
 - Global Monitoring Nodes
 - Enterprise Plans
-

14. Product Vision Summary

DevAlive is a modern SaaS platform built specifically for developers who rely on free-tier hosting platforms.

The platform provides automated monitoring, uptime tracking, alerting, and reliability analytics to ensure projects remain accessible, professional, and recruiter-ready.

By focusing on simplicity, automation, and developer needs, DevAlive aims to become the preferred monitoring platform for students, freelancers, indie hackers, and software engineers worldwide.